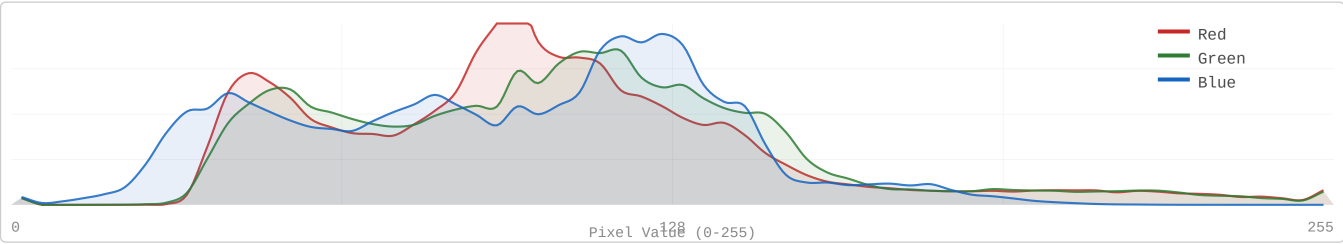
Decorrelation Gap: KODIM16

768x512 | 24-bit RGB | PNG (lossless)

Aerial View of City and Harbor | Strongly one-dimensional (PC1 = 96.4%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

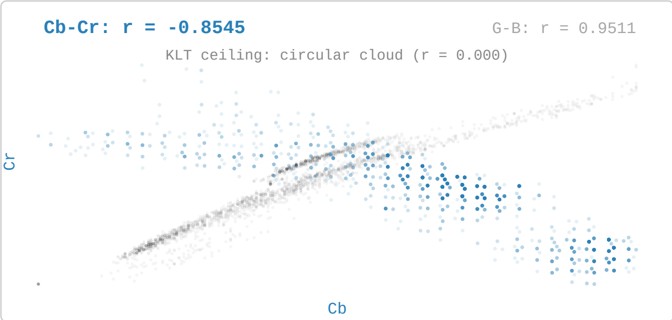
	R	G	B
R	1.0000	0.9842	0.8923
G	0.9842	1.0000	0.9511
B	0.8923	0.9511	1.0000

Cb-Cr:
-0.854
(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.3919	0.0376
Cb	-0.3919	1.0000	-0.8545
Cr	0.0376	-0.8545	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.984232	-0.391925	0.3919
R-B → Y-Cr	0.892330	0.037581	0.0376
G-B → Cb-Cr	0.951070	-0.854464	0.8545
Avg |r|	0.942544	0.427990	0.4280

Decorrelation Gap: KODIM16

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

54.6% Moderate

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

7.5%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	48.64 dB
G	51.02 dB
B	45.68 dB
Overall	47.91 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5939	-0.6071	-0.5278	5240.94
PC2	0.6045	0.0962	-0.7908	189.20
PC3	-0.5309	0.7887	-0.3099	6.36

8. PCA Derived Metrics

Condition Number

824.20

PC1/PC2: 27.70 | PC2/PC3: 29.75

Blue Channel Independence

7.5%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.61)

9. Redundancy Profile

Classification: Strongly one-dimensional | CN: 824.20 | PC1: 96.4%

RGB Correlation: Avg |r| = 0.942544 | R-G: 0.9842 | R-B: 0.8923 | G-B: 0.9511

YCbCr Residual: Avg |r| = 0.427990 | Y-Cb: -0.3919 | Y-Cr: 0.0376 | Cb-Cr: -0.8545

Efficiency: 54.6% of KLT ceiling | Gap: 0.4280 avg |r| remaining

Blue Channel: 7.5% independent | 4:2:0 PSNR: 47.91 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.